HOME LAB NETWORK

SWITCH VLAN CONFIGURATION

Netgear GS308E v4 — 802.1Q VLAN Port Assignments

Router-on-a-Stick Topology | pfSense 2.8.1 | Protectli FW6E

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1. Architecture Overview

This document details the 802.1Q VLAN configuration of the Netgear GS308E v4 managed switch in a router-on-a-stick topology. All inter-VLAN routing is handled by pfSense through a single trunk link on switch port 1. End devices connect to access ports that are untagged on a single VLAN, providing transparent network segmentation.

1.1 Topology Design

Router-on-a-Stick:

A single physical cable connects the switch (port 1) to pfSense (igb1/LAN). This cable carries all VLAN traffic using 802.1Q tags. pfSense receives tagged frames, routes them through the appropriate VLAN interface, applies firewall rules, and sends them to the VPN or blocks them. This eliminates the need for a dedicated physical port per VLAN on the firewall.

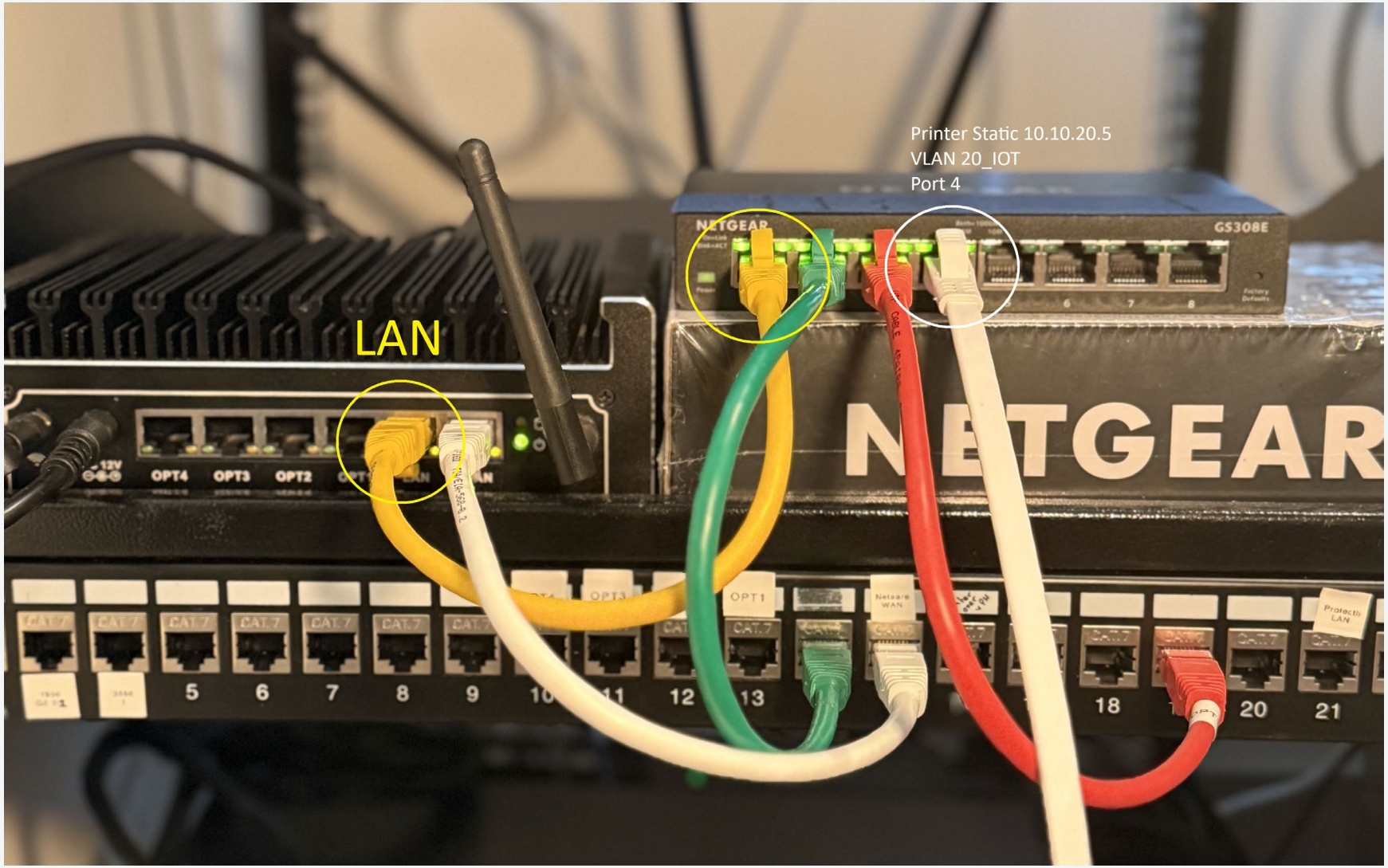
Traffic Flow Example — PC on port 2 browsing the internet:

1. PC sends untagged Ethernet frame to switch port 2
2. Switch assigns PVID 10 → frame tagged as VLAN 10 internally
3. Switch forwards tagged frame out port 1 (trunk) to pfSense
4. pfSense receives on igb1, reads VLAN 10 tag, routes to VLAN10_USERS interface
5. Firewall rules evaluated → traffic sent through VPN_FAILOVER gateway
6. Response returns same path in reverse

Traffic Flow Example — PC printing to printer on VLAN 20:

1. PC (port 2, VLAN 10) sends print job to 10.10.20.5
2. Switch tags as VLAN 10, sends to pfSense via trunk
3. pfSense evaluates VLAN10_USERS firewall rules → printer pass rule matches (port 9100 to 10.10.20.5)
4. pfSense routes packet to VLAN20_IOT interface, tags as VLAN 20
5. Switch receives VLAN 20 tagged frame on port 1, strips tag, forwards to port 4 (printer)
6. Printer receives untagged frame and processes print job

Rationale: Router-on-a-stick uses fewer physical ports (only 1 uplink needed) and centralizes all routing/security decisions at the firewall. The tradeoff is bandwidth — all inter-VLAN traffic shares one gigabit link. For a home lab with under 10 devices, this is more than sufficient.

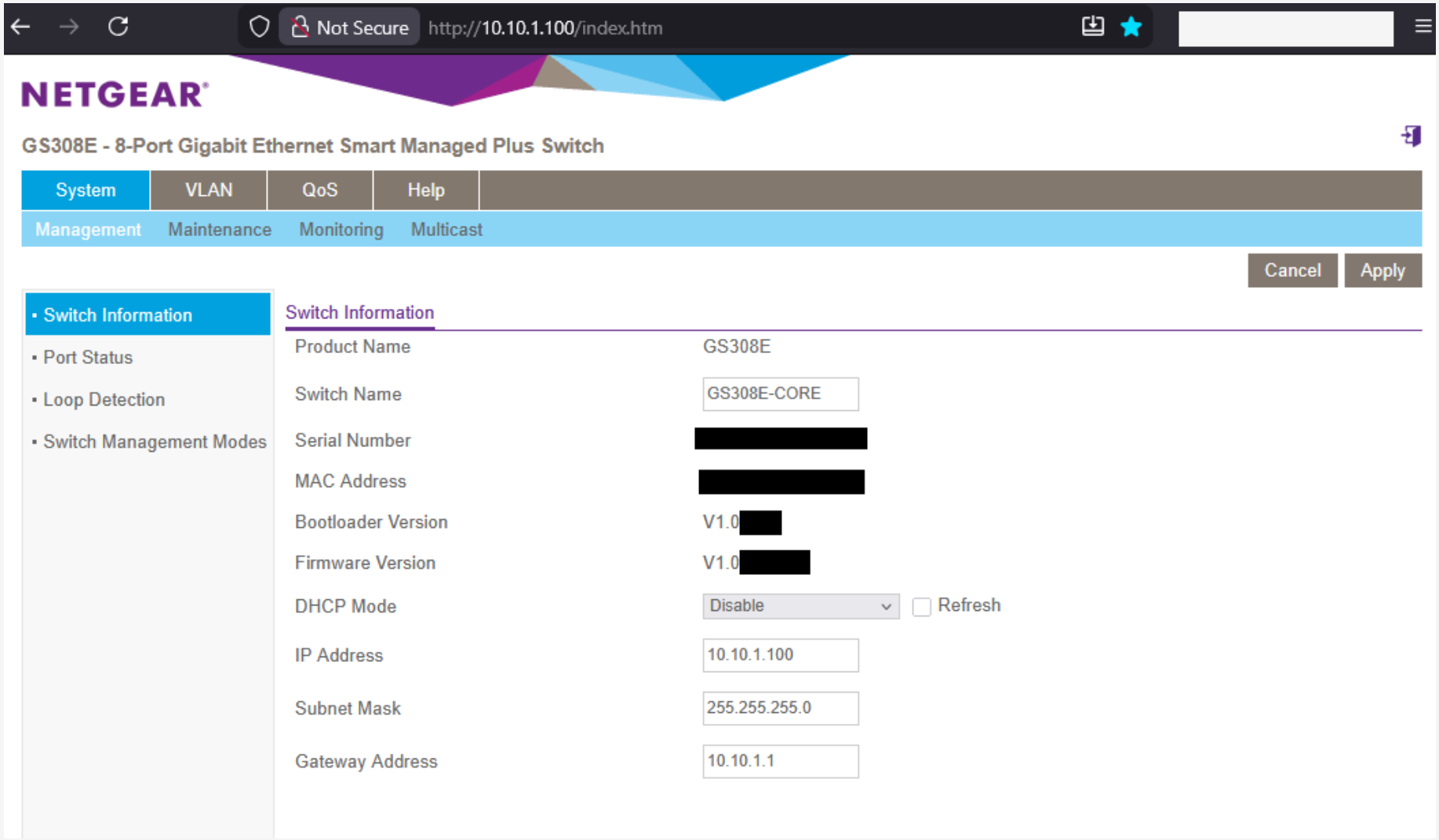


Physical topology diagram: Protectli FW6E → Port 1 GS308E → Ports 2-8 to devices

2. Switch Hardware Details

Property	Value
Model	Netgear GS308E v4 — 8-Port Gigabit Ethernet Smart Managed Plus Switch
MAC Address	xx:xx:xx:xx:xx:xx
Serial Number	xxxxxxxxxxxxx
Firmware	V1.xx.xxxx
Management IP	10.10.1.100 (VLAN 1 — Native LAN)
Subnet Mask	255.255.255.0
Gateway	10.10.1.1 (pfSense LAN interface)
802.1Q VLAN Mode	Advanced (Enabled)
Factory Reset IP	192.168.0.239 (emergency access: set PC to 192.168.0.100/24)
Uplink Port	Port 1 → pfSense igb1 (LAN)

⚠ **Note:** Emergency access: If the switch becomes unreachable at 10.10.1.100, set your PC to static IP 192.168.0.100/24, connect directly to any switch port, and browse to 192.168.0.239 (factory default IP). This bypasses all VLAN config.



GS308E System > Management > System Information page

3. VLAN Summary

Six VLANs are configured to segment the network by function and security level. Each VLAN maps to a pfSense interface with dedicated firewall rules, DHCP scope, and routing policy.

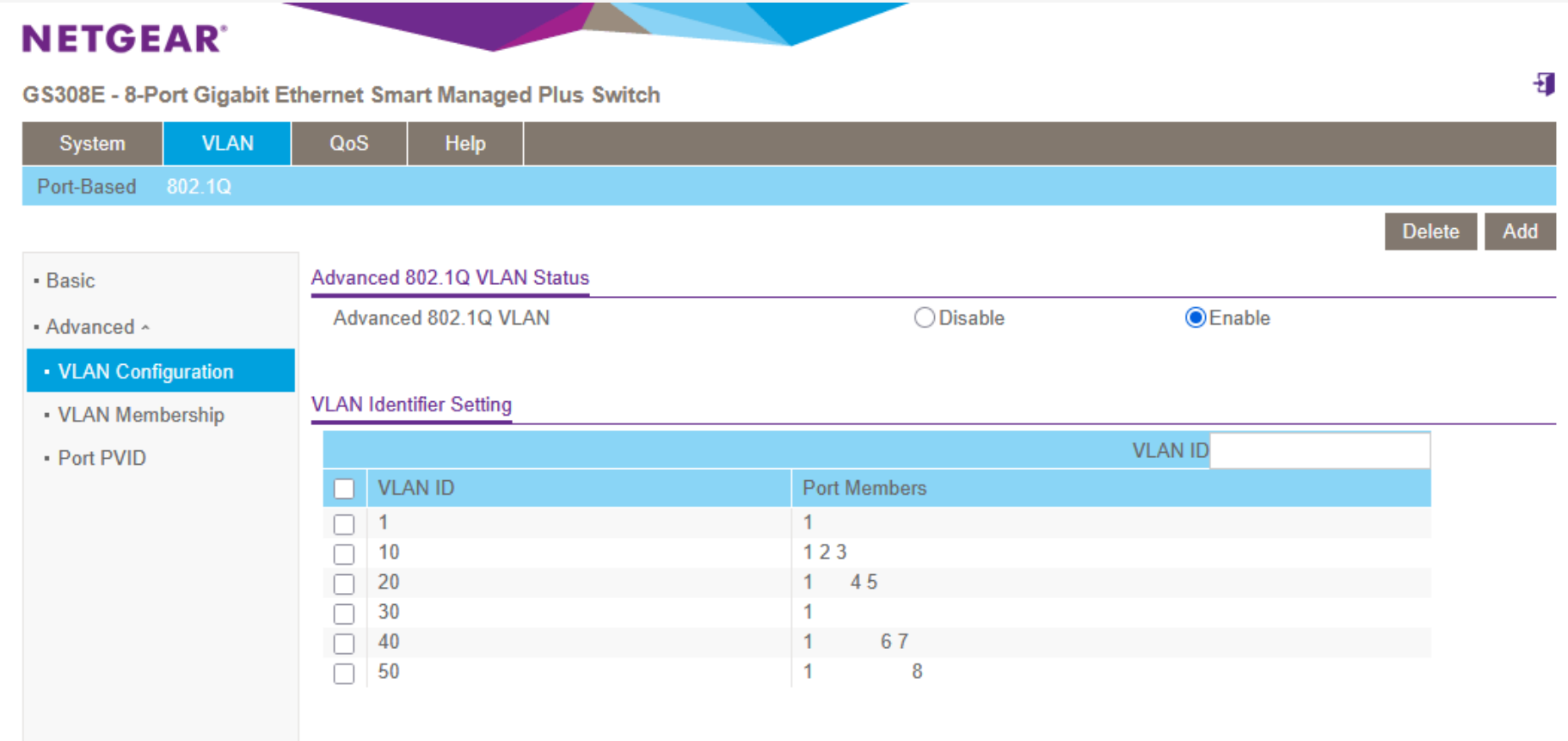
VLAN ID	Name	Subnet	Gateway	VPN Routed	Purpose
1	Native LAN	10.10.1.0/24	10.10.1.1	Yes	Management / Switch access
10	VLAN10_USERS	10.10.10.0/24	10.10.10.1	Yes	Workstations / Personal devices
20	VLAN20_IOT	10.10.20.0/24	10.10.20.1	Yes	IoT devices / Printer
30	VLAN30_GUEST	10.10.30.0/24	10.10.30.1	Yes	Guest Wi-Fi (future)
40	VLAN40_LAB	10.10.40.0/24	10.10.40.1	Yes	Lab gear (Cisco 3560, 1900)
50	VLAN50_MGMT	10.10.50.0/24	10.10.50.1	No	Network management (break glass)

Rationale: Network segmentation isolates device classes by trust level. User workstations (VLAN 10) are separated from IoT devices (VLAN 20) which are inherently less secure. Guest devices (VLAN 30) have zero access to internal resources. Lab equipment (VLAN 40) runs experimental configs that could be vulnerable. Management (VLAN 50) is the break-glass network with no VPN dependency.

4. Port Assignments

Each physical port is assigned to exactly one VLAN as an access port, except port 1 which is the trunk carrying all VLANs to pfSense.

Port	Device	VLAN	Tagging	PVID	Role	Notes
1	pfSense LAN (igb1)	All	**Trunk**	1	Uplink	Tagged: 10, 20, 30, 40, 50. Untagged: 1 (native). All VLAN traffic traverses this single cable to pfSense.
2	Primary Workstation	10	Untagged	10	Access	PC connects here. Traffic auto-tagged as VLAN 10 by switch. Device unaware of VLAN.
3	Netgear R6400 AP	10	Untagged	10	Access	Wi-Fi AP in AP-only mode. Serves VLAN 10 wireless clients. Router features disabled.
4	Network Printer	20	Untagged	20	Access	Printer isolated on IoT VLAN. Static IP: 10.10.20.5 via DHCP reservation. Accessible from VLAN10 via firewall rule (port 9100).
5	IoT Devices	20	Untagged	20	Access	Reserved for future IoT devices. Same isolation as port 4.
6	Lab Gear	40	Untagged	40	Access	Cisco Catalyst 3560 or other lab equipment. Fully isolated from production.
7	Lab Gear	40	Untagged	40	Access	Cisco 1900 or additional lab equipment. Same isolation as port 6.
8	MGMT Spare	50	Untagged	50	Access	Break-glass management port. No VPN, direct WAN access. Use only for emergency network administration.



GS308E VLAN > 802.1Q > VLAN Configuration page showing all VLANs and port members

5. VLAN Membership Matrix

This matrix shows the tagging mode for each port on each VLAN. T = Tagged (802.1Q header present), U = Untagged (header stripped), — = Not a member.

VLAN	Port 1	Port 2	Port 3	Port 4	Port 5	Port 6	Port 7	Port 8
1	U	—	—	—	—	—	—	—
10	T	U	U	—	—	—	—	—
20	T	—	—	U	U	—	—	—
30	T	—	—	—	—	—	—	—
40	T	—	—	—	—	U	U	—
50	T	—	—	—	—	—	—	U

5.1 Tagging Explained

Tagged (T) — Trunk mode:

The switch adds an 802.1Q header containing the VLAN ID to every frame sent out this port. The receiving device (pfSense) reads the tag to determine which VLAN the frame belongs to. Port 1 is tagged for VLANs 10, 20, 30, 40, and 50 because pfSense needs to distinguish traffic from all VLANs on a single cable.

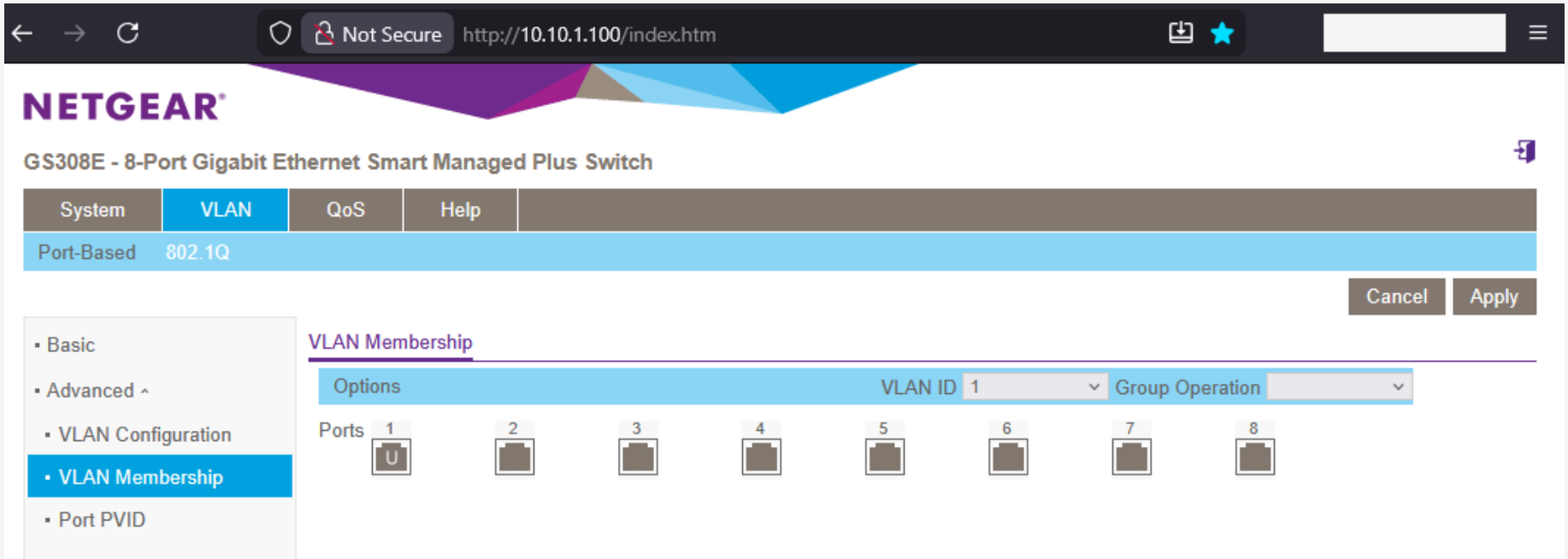
Untagged (U) — Access mode:

The switch strips the 802.1Q header before sending the frame to the device. The end device (PC, printer, AP) sees a normal Ethernet frame with no VLAN awareness required. When the device sends traffic back, the switch assigns it to the VLAN defined by the port's PVID.

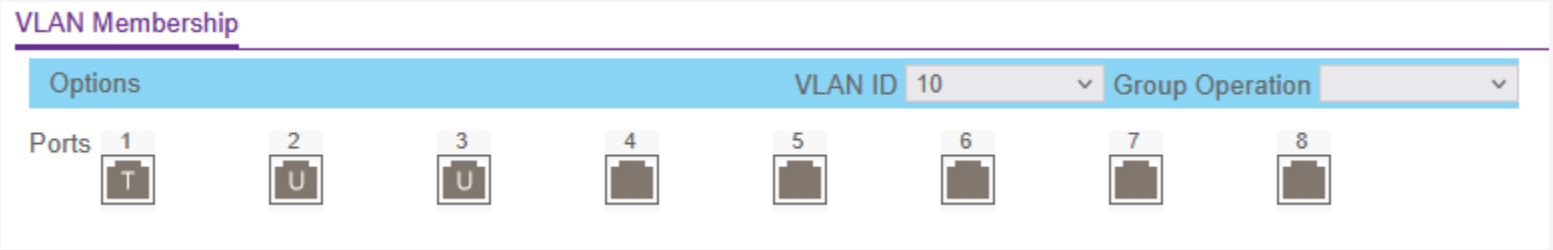
Not a member (—):

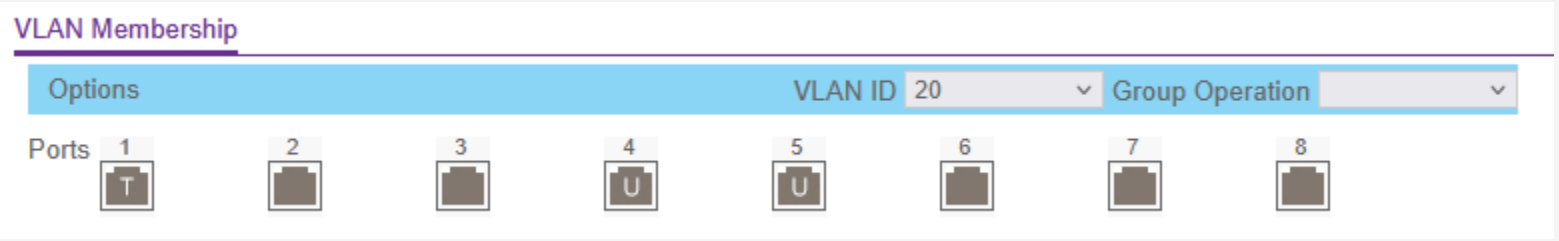
The port has no association with this VLAN. Traffic for this VLAN will never be forwarded to this port, and traffic from this port will never be tagged with this VLAN ID. This enforces hard isolation between segments.

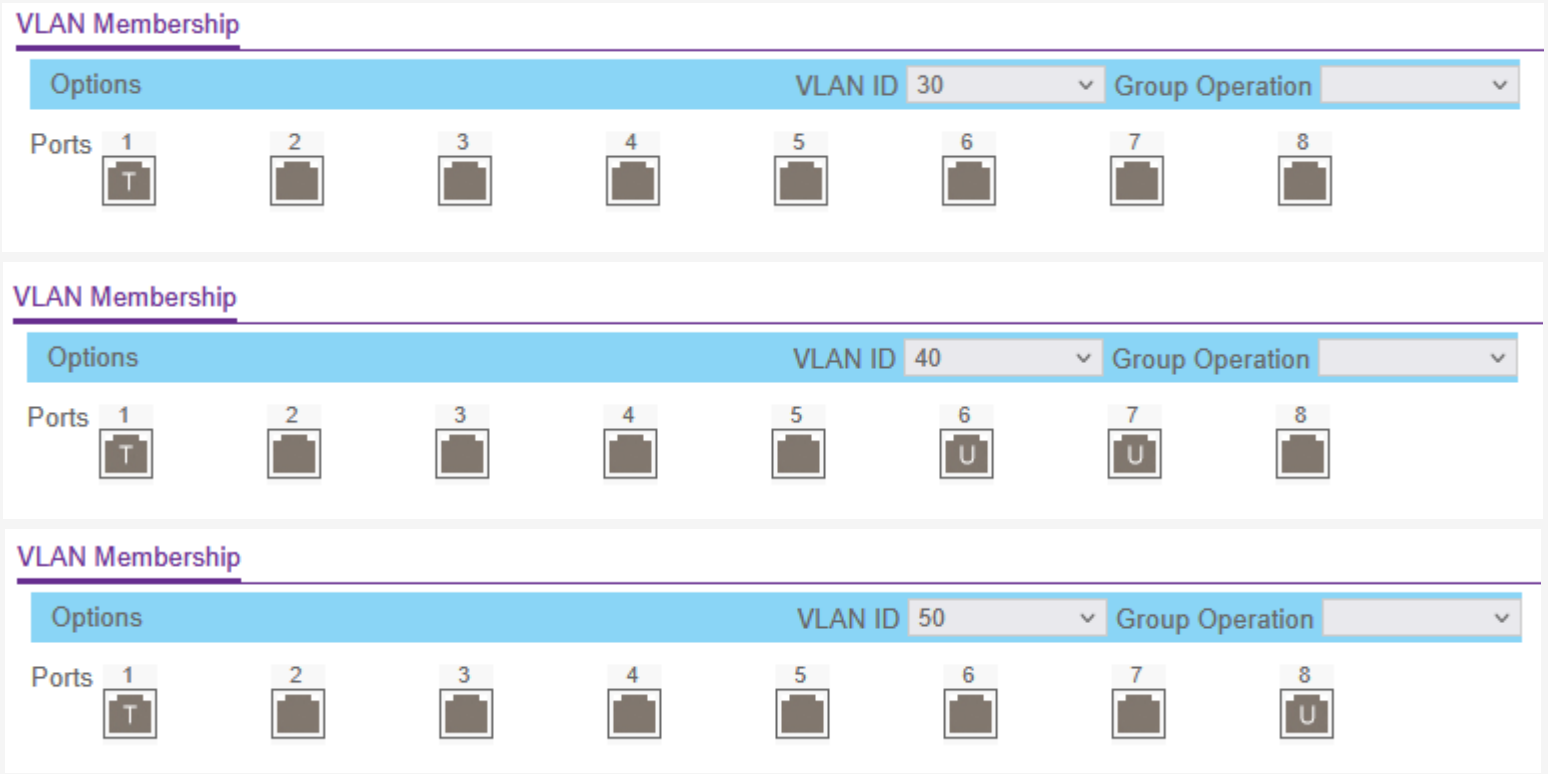
Key principle: Each access port is Untagged on exactly one VLAN and not a member of any other. This ensures a device can only communicate on its assigned VLAN. The only exception is port 1 (trunk) which is a member of all VLANs.



GS308E VLAN > 802.1Q > VLAN Membership page — cycle through each VLAN ID





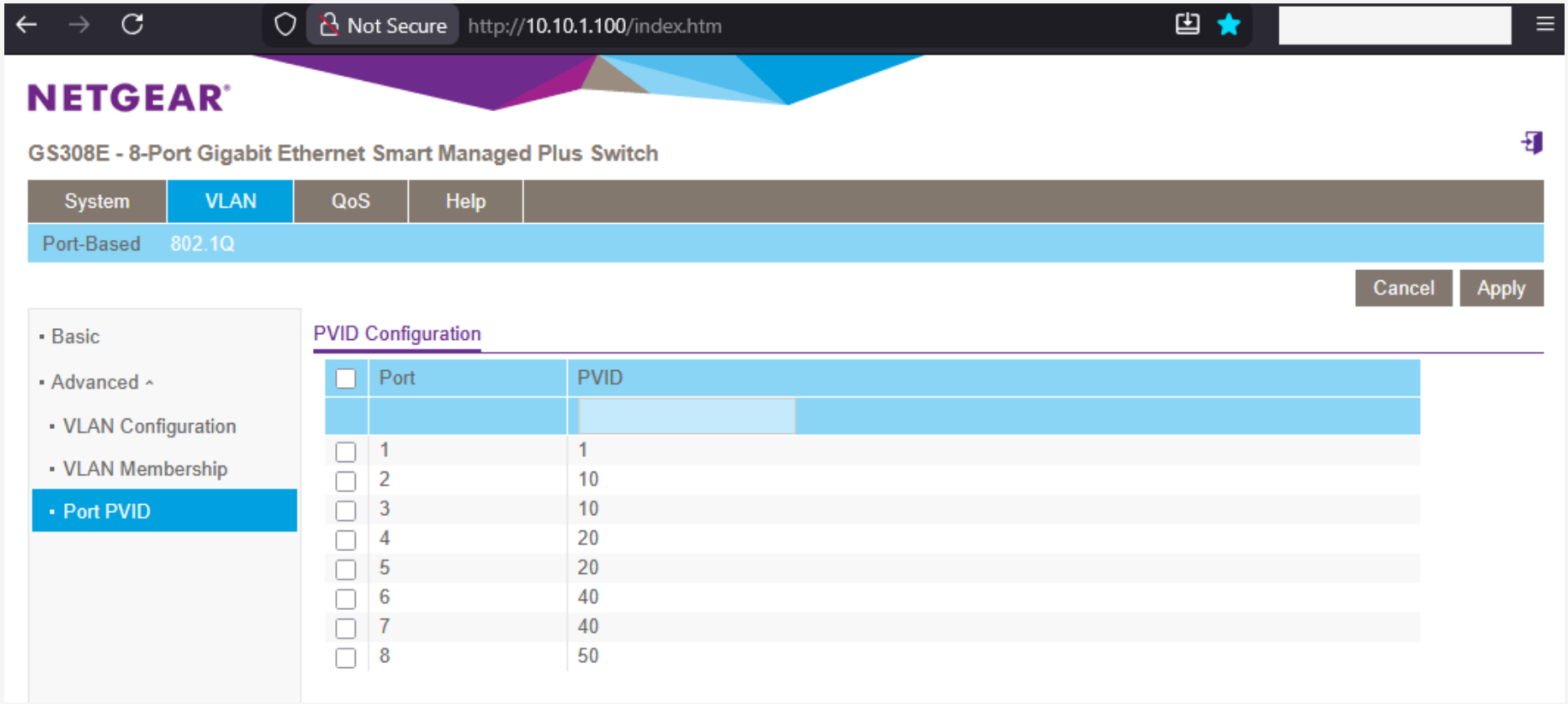

VLAN Membership
Options
VLAN ID 30
Group Operation
Ports 1 2 3 4 5 6 7 8
T
VLAN Membership
Options
VLAN ID 40
Group Operation
Ports 1 2 3 4 5 6 7 8
T U U
VLAN Membership
Options
VLAN ID 50
Group Operation
Ports 1 2 3 4 5 6 7 8
T U

6. Port PVID Configuration

PVID (Port VLAN ID) determines which VLAN is assigned to untagged ingress traffic. When a device sends a frame without an 802.1Q tag, the switch assigns the frame to the VLAN matching the port's PVID.

Port	PVID	VLAN Name	Subnet	Effect
1	1	Native LAN	10.10.1.0/24	Untagged ingress traffic assigned to management VLAN
2	10	VLAN10_USERS	10.10.10.0/24	PC traffic auto-tagged as VLAN 10
3	10	VLAN10_USERS	10.10.10.0/24	AP traffic auto-tagged as VLAN 10
4	20	VLAN20_IOT	10.10.20.0/24	Printer traffic auto-tagged as VLAN 20
5	20	VLAN20_IOT	10.10.20.0/24	IoT device traffic auto-tagged as VLAN 20
6	40	VLAN40_LAB	10.10.40.0/24	Lab gear traffic auto-tagged as VLAN 40
7	40	VLAN40_LAB	10.10.40.0/24	Lab gear traffic auto-tagged as VLAN 40
8	50	VLAN50_MGMT	10.10.50.0/24	Management traffic auto-tagged as VLAN 50

Rationale: PVID must match the port's untagged VLAN membership. If port 4 is untagged on VLAN 20, its PVID must be 20. A mismatch causes traffic to be assigned to the wrong VLAN or dropped entirely. The GS308E enforces this — it will not allow a PVID that doesn't match a VLAN membership.



GS308E VLAN > 802.1Q > Port PVID page showing all 8 port PVID assignments

7. Verification Procedures

7.1 Connectivity Test per VLAN

For each VLAN, connect a device to the assigned port and verify:

1. Device receives correct IP from DHCP (e.g., 10.10.10.x on port 2, 10.10.20.x on port 4)
2. Device can ping its gateway (e.g., 10.10.10.1 from VLAN 10)
3. Device can resolve DNS (nslookup google.com [gateway IP])
4. Device can browse the internet
5. Device CANNOT ping other VLAN gateways (e.g., 10.10.20.1 from VLAN 10 should fail)

7.2 Isolation Test

From a device on VLAN 10 (port 2):

- ping 10.10.20.1 → should FAIL (blocked by RFC1918 rule)
- ping 10.10.30.1 → should FAIL
- ping 10.10.40.1 → should FAIL
- ping 10.10.50.1 → should FAIL
- ping 10.10.20.5 port 9100 → should SUCCEED (printer exception rule)

7.3 Trunk Verification

From pfSense, verify all VLAN interfaces show "up" under Status > Interfaces:

- VLAN10_USERS (igb1.10) — up
- VLAN20_IOT (igb1.20) — up
- VLAN30_GUEST (igb1.30) — up
- VLAN40_LAB (igb1.40) — up
- VLAN50_MGMT (igb1.50) — up

Interfaces

WAN	↑	1000baseT <full-duplex>	[redacted]8
LAN	↑	1000baseT <full-duplex>	10.10.1.1
VPN_CHI	↑		[redacted]5
VLAN10_USERS	↑	1000baseT <full-duplex>	10.10.10.1
VLAN20_IOT	↑	1000baseT <full-duplex>	10.10.20.1
VLAN30_GUEST	↑	1000baseT <full-duplex>	10.10.30.1
VLAN40_LAB	↑	1000baseT <full-duplex>	10.10.40.1
VLAN50_MGMT	↑	1000baseT <full-duplex>	10.10.50.1
VPN_NYC	↑		[redacted]5

pfSense Status > Interfaces showing all VLAN interfaces up

8. Troubleshooting

8.1 Device Gets Wrong IP

Symptom: Device on port 4 gets 10.10.10.x instead of 10.10.20.x

Cause: PVID is set to 10 instead of 20, or port is still a member of VLAN 10

Fix: Check Port PVID (must be 20) and VLAN Membership (must be Untagged on VLAN 20 only)

8.2 Device Gets No IP

Symptom: Device shows 169.xxx.x.x (APIPA) or no connectivity

Cause: Port is not a member of any VLAN, or DHCP server is not running on the corresponding VLAN interface in pfSense

Fix: Verify VLAN membership, verify DHCP is enabled on the correct pfSense VLAN interface

8.3 Cannot Access Switch Management

Symptom: Cannot browse to 10.10.1.100

Cause: Your device is on a different VLAN (e.g., VLAN 10) and inter-VLAN traffic to 10.10.1.x is blocked by RFC1918 rule

Fix: Use the TEMP firewall rule on VLAN10_USERS (port 80/443 to 10.10.1.100) or connect to a native LAN port

8.4 Emergency Recovery

If all access is lost:

1. Set PC to static IP: 192.168.0.100, subnet 255.255.255.0
2. Connect PC directly to any switch port
3. Browse to 192.168.0.239 (factory default)
4. Reconfigure VLANs from scratch

⚠ **Note:** Factory reset erases ALL switch configuration. Only use as a last resort. Always take a backup of the switch config (System > Maintenance > Export) before making changes.

9. Future Planning

9.1 Planned Changes

- Configure R6400 AP with multiple SSIDs mapped to VLANs (if supported) or dedicate to VLAN 10
- Remove TEMP switch management firewall rule from VLAN10_USERS once switch config is finalized
- Consider adding a dedicated VLAN 30 (Guest) access port if wired guest access is needed

9.2 Scaling Considerations

- The GS308E has 8 ports. If more ports are needed, add a second GS308E or upgrade to a 16/24-port managed switch
- Second switch would connect via a trunk port, mirroring the VLAN configuration
- For higher throughput between VLANs, consider a Layer 3 switch (e.g., Cisco Catalyst 3560) to offload inter-VLAN routing from pfSense

9.3 Configuration Change Procedure

1. Document the proposed port/VLAN change in this document
2. Export switch configuration (System > Maintenance > Export)
3. Make the change on the switch
4. Test connectivity from the affected port
5. Verify isolation from other VLANs
6. Update this document with the final configuration and screenshots
7. Take a pfSense backup if firewall rules were also modified